



Student No:	

Mathematics Extension 1

Year 12, Assessment task 3, Term 2 2023

General Instruction:

- Reading time – 10 minutes
- Working time – 120 minutes
- Write using black/blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:

Section I – 10 marks (pages 1–3)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 4–7)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Class Teacher (please tick one name)

- ☐ Mr Berry
☐ Ms Fu
☐ Mr Ireland
☐ Ms Lee
☐ Mr Lin
☐ Mr Umakanthan
☐ Dr Vranešević

Question No.	1-10	11	12	13	14	Total	Total %
Marks	/10	/15	/15	/15	/15	/70	

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Section I**10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

1. Which of the following is the correct expression for $\int \frac{dx}{\sqrt{36-x^2}}$?

A) $\arccos \frac{x}{6} + c$;

B) $\arccos 6x + c$;

C) $\arcsin \frac{x}{6} + c$;

D) $\arcsin 6x + c$.

2. Find the fifth term in the expansion of $\left(3x^2 - \frac{1}{2x}\right)^8$.

A) $\binom{8}{4} \left(\frac{3}{2}\right)^4 x^4$;

B) $\binom{4}{8} \left(\frac{2}{3}\right)^4 x^{-4}$;

C) $\binom{8}{4} \left(\frac{3}{2}\right)^4 x^{-4}$;

D) $\binom{4}{8} \left(\frac{2}{3}\right)^4 x^4$.

3. A projectile has an initial velocity vector $\vec{v} = \begin{pmatrix} 2\sqrt{3} \\ 2 \end{pmatrix}$. Which of the following is the correct statement of its initial speed and angle of projection from the horizontal?

A) 4 m/s at 30° ;

B) $\sqrt{10}$ m/s at 30° ;

C) 4 m/s at 60° ;

D) $\sqrt{10}$ m/s at 60° .

4. If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?

A) $f^{-1}(x) = e^{x-2}$;

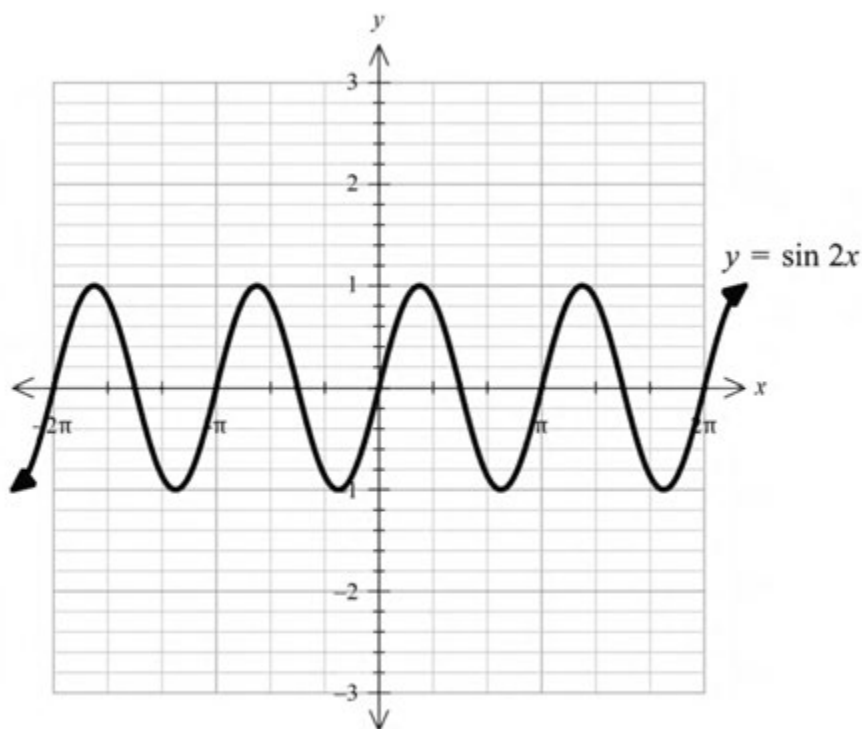
B) $f^{-1}(x) = e^{x+2}$;

C) $f^{-1}(x) = \log_e x - 2$;

D) $f^{-1}(x) = \log_e x + 2$.

5. The radius of a sphere is increasing at the rate of 5 centimetres per minute. What is the rate of increase of the volume of the sphere, in cubic centimetres per minute, when the radius is 6 centimetres?
- A) 144π ;
B) 288π ;
C) 600π ;
D) 720π .
6. If $x + a$ is a factor of $4x^3 + 13x^2 + ax$, then the value of a is:
- A) 3;
B) 1;
C) -3 ;
D) -4 .
7. If $\vec{a} = -2\hat{i} - \hat{j}$ and $\vec{b} = m\hat{i} - \hat{j}$, where m is a real constant, then the vector $\vec{a} - \vec{b}$ will be perpendicular to vector $\vec{b}$ when m is:
- A) 0 or 2;
B) 0 or -2 ;
C) $-1 \pm \sqrt{3}$;
D) $-1 \pm \sqrt{5}$.
8. If $t = \tan \frac{\theta}{2}$ which of the following expressions is equivalent to $4 \sin \theta + 3 \cos \theta + 5$?
- A) $\frac{2(t+2)^2}{1-t^2}$;
B) $\frac{(t+4)^2}{1-t^2}$;
C) $\frac{2(t+2)^2}{1+t^2}$;
D) $\frac{(t+4)^2}{1+t^2}$.
9. What is the value of $\cos^{-1}(\sin \alpha)$, where $\frac{\pi}{2} < \alpha < \pi$?
- A) $\pi - \alpha$;
B) $\frac{\pi}{2} - \alpha$;
C) $\alpha - \frac{\pi}{2}$;
D) $\alpha + \frac{\pi}{2}$.

10. The function $y = \sin 2x$ is shown in the diagram.



If this function is transformed using steps I, II and III as below:

I.	Reflected about the x -axis
II.	Vertically translated 1 unit down
III.	Dilated horizontally by a scale factor of 2.

Which equation would represent the transformed function?

- A) $y = -(\sin x + 1)$;
- B) $x = 2(\sin(2y) - 1)$;
- C) $y = -(\sin(4x)) + 2$;
- D) $y = -2 \sin(2x) - 1$.

End of Section I

Section II**60 marks****Attempt Questions 11–14****Allow about 1 hour and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 marks) Use the Question 11 Writing Booklet(a) Solve for x :

$$\frac{1-2x}{1+x} \geq 1. \quad 3$$

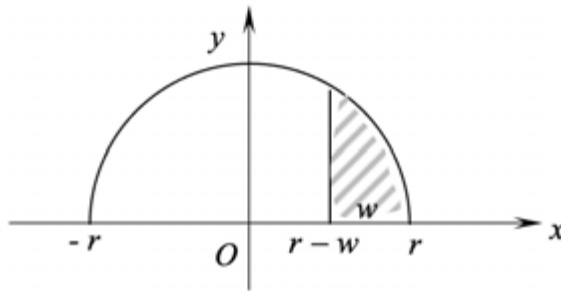
(b)

(i) Express $\sin 2x + \sqrt{3} \cos 2x$ in the form of $R \sin(2x + \alpha)$. 2(ii) Find the greatest value of $\sin 2x + \sqrt{3} \cos 2x$ and the values of x , $0 \leq x \leq 2\pi$, at which it occurs. 2(iii) Hence, or otherwise, solve $\sin 2x + \sqrt{3} \cos 2x = 1$, $0 \leq x \leq 2\pi$ 3(c) By using substitution $u = \tan^{-1} x$ evaluate $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$ 2(d) $P(x)$ is an odd polynomial of degree 3. It has $(x+4)$ as a factor and, when it is divided by $(x-3)$, the remainder is 21. Find $P(x)$. 3

Question 12. (15 marks) Use the Question 12 Writing Booklet

- (a) The semi-circle $y = \sqrt{r^2 - x^2}$ is shown on diagram.

2



The shaded area of thickness w is rotated about the x -axis to form the volume of a 'cap'. Show that the volume of the solid of revolution V is given by

$$V = \frac{\pi}{3}(3r - w)w^2.$$

- (b) Consider the equation $\cos x - 2\sin x = 1$, for $-\pi \leq x \leq \pi$.

- (i) Show that the equation can be written as $t^2 + 2t = 0$, where $t = \tan \frac{x}{2}$.

2

- (ii) Hence, solve the equation for $-\pi \leq x \leq \pi$, giving solutions to the 2 decimal places where necessary.

2

- (c) (i) By equating coefficients, find the values of P and Q in the identity

$$P(2\sin x + \cos x) + Q(2\cos x - \sin x) \equiv 7\sin x + 11\cos x$$

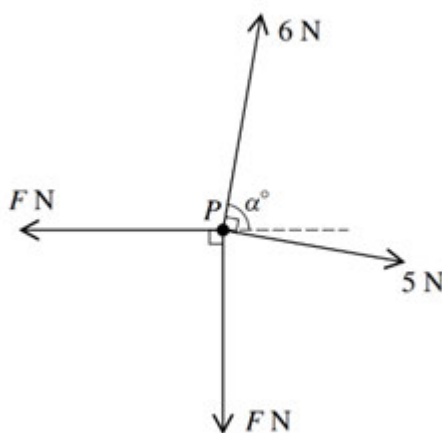
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- (ii) Hence, or otherwise, evaluate

$$\int_0^{\frac{\pi}{2}} \frac{7\sin x + 11\cos x}{2\sin x + \cos x} dx$$

2

- (d)



A particle P is in equilibrium under the action of four forces of magnitude 6 N, 5 N, F N and F N acting in the directions shown in the diagram on the right.

- (i) By considering the vertical components of the forces, find an expression for F in terms of α .

2

- (ii) Hence, determine the value of α , correct to the nearest degree.

3

Question 13. (15 marks) Use the Question 13 Writing Booklet

- (a) (i) Prove that

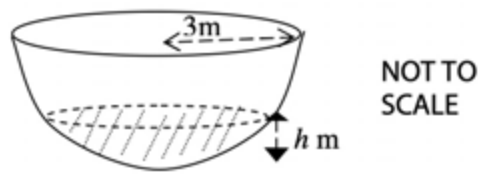
$$\cos^{-1}(-x) = \pi - \cos^{-1}x \quad 2$$

- (iii) Hence, or otherwise, evaluate

$$\cos^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) \quad 2$$

- (b)

A hemi-spherical bowl has a radius of 3 m. Oil is poured in at a constant rate of $\frac{\pi}{3}$ m³/min.



- (i) Show that, when the depth of the oil is
- h
- metres, the volume of oil is:

$$V = \frac{\pi}{3}(9h^2 - h^3) \quad 2$$

- (ii) Show that the depth of the oil in the bowl after 8 minutes is 1 meter?

3

- (iii) At what rate is
- h
- increasing at this time?

2

- (c) Prove, by the method of mathematical induction that:

$$\sin q + \sin 3q + \sin 5q + \dots + \sin(2n-1)q = \frac{1 - \cos 2nq}{2 \sin q}, \text{ for } n = 1, 2, 3, \dots \quad 4$$

Question 14. (15 marks) Use the Question 14 Writing Booklet

- (a) If
- $f(x) = 2 - \sqrt{x}$
- ,
- $x \geq 0$
- , and
- $g(x) = (x - 2)^2$
- for all
- x
- , find the values of
- x
- for which

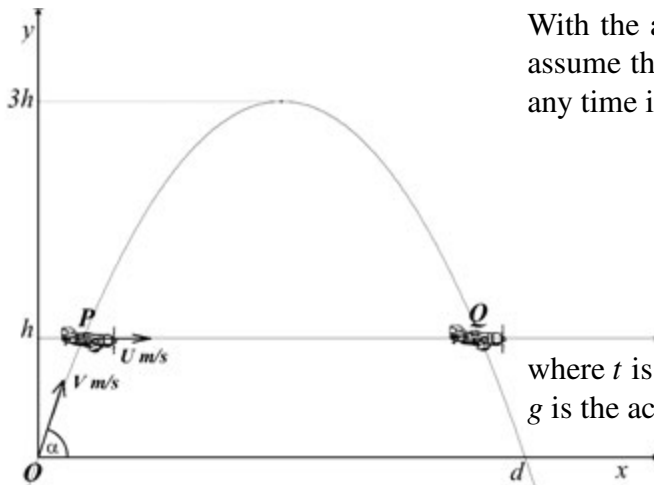
$$f(g(x)) = g(f(x))$$

3

- (b) An enemy plane is flying horizontally at height
- h
- metres with speed
- U
- m/s.

When it is at point P, a ground rocket is fired towards it from the origin O with speed V m/s and angle of elevation α .

The rocket misses the plane, passing too late through the point P. However, it goes on to reach a maximum height of $3h$ metres and then on its descent strikes the plane at Q.



With the axes as shown in the diagram, you may assume that the position vector, $\vec{s}$, of the rocket at any time is given by:

$$\vec{s}(t) = x(t)\hat{i} + y(t)\hat{j},$$

$$x(t) = vt \cos \alpha$$

$$y(t) = -\frac{1}{2}gt^2 + vt \sin \alpha$$

where t is the time in seconds after firing and g is the acceleration due to gravity.

- (i) Show that the initial vertical velocity component (
- $v \sin \alpha$
-) of the rocket's speed equals

$$\sqrt{6gh}.$$

2

- (ii) If the rocket had not struck the plane at Q, it would have returned to the x-axis at a distance
- d
- metres from O. Show that the horizontal component (
- $v \cos \alpha$
-) of the rocket's speed equals

$$\frac{gd}{2\sqrt{6gh}}$$

2

- (iii) Show that Cartesian equation of the trajectory of the rocket is

$$y = \frac{12hx}{d} \left(1 - \frac{x}{d}\right).$$

2

- (iv) If the horizontal component of the rocket's speed is
- $100(3 + \sqrt{6})$
- m/s, find the time taken by the rocket to strike the plane at Q, in terms of
- d
- .

2

- (v) Find the speed of the enemy plane.

1

- (c) Given the binomial expansion of:

$$(1+x)^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

$$(1+x)^{n+1} = b_0 + b_1x + b_2x^2 + \dots + b_{n+1}x^{n+1}$$

and knowing $(1+x)^{n+1} = (1+x)^n(1+x)$, find the expression, **in terms of n only**, of:

$$\frac{1}{a_0a_1a_2\dots a_n} \times (a_0 + a_1)(a_1 + a_2)\dots(a_{n-1} + a_n) \text{ for } n = 1, 2, 3, \dots$$

3

End of Examination

MC

$$Q_1 \int \frac{dx}{\sqrt{36-x^2}} = \sin^{-1} \frac{x}{6} + C \quad \text{or} \quad \arcsin \frac{x}{6} + C \Rightarrow \textcircled{C}$$

$$Q_2 (3x^2 - \frac{1}{2}x)^8, T_{r+1} = \binom{n}{r} \underbrace{I^{n-r}}_{1. \text{ term}} \underbrace{II^r}_{2. \text{ term}}$$

$$T_5 = \binom{8}{4} (3x^2)^{8-4} \left(-\frac{1}{2}x\right)^4$$

$$= \binom{8}{4} \frac{3^4}{2^4} (-1)^4 x^8 x^4$$

$$= \binom{8}{4} \left(\frac{3}{2}\right)^4 (-1)^4 x^4$$

$$= \binom{8}{4} \left(\frac{3}{2}\right)^4 x^4 \Rightarrow \textcircled{A}$$

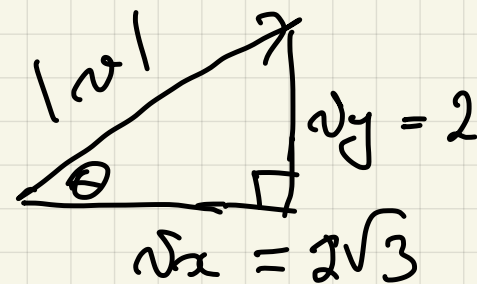
$$Q_3 \vec{w} = \begin{pmatrix} 2\sqrt{3} \\ 2 \end{pmatrix}$$

$$|\vec{w}|^2 = (2\sqrt{3})^2 + 2^2$$

$$= 4 \times 3 + 4$$

$$= 16$$

$$\therefore |\vec{w}| = 4$$



$$\tan \theta = \frac{2}{2\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$\theta = 30^\circ \Rightarrow \textcircled{A}$$

Q4. $f(x) = e^{x+2}$ $f^{-1}(x) = ?$

$$x = e^{y+2}$$

$$\ln x = y + 2$$

$$y = \ln x - 2 \Rightarrow \textcircled{C}$$

Q6. $P(x) = 4x^3 + 13x^2 + ax$ A
 $P(-a) = 0$ $(x+a) = (x-(-a))$

$$= -4a^3 + 13a^2 - a^2 = 0$$

$$\Rightarrow -4a^3 + 12a^2 = 0$$

$$-4a^2(a-3) = 0$$

$$\therefore a = 3 \Rightarrow \textcircled{A}$$

$$\vec{b} = \begin{pmatrix} m \\ -1 \end{pmatrix}$$

Q7. $\vec{a} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$

$$\vec{a} - \vec{b} = (-2-m)\hat{i} + (-1+m)\hat{j} = \begin{pmatrix} -2-m \\ 0 \end{pmatrix}$$

$$(\vec{a} - \vec{b}) \perp \vec{b}, \text{ if } (\vec{a} - \vec{b}) \cdot \vec{b} = 0$$

$$(-2-m) \cdot m + (-1) \cdot 0 = 0$$

$$-2m - m^2 = 0$$

$$m(2+m) = 0$$

$$\Rightarrow m = 0 \text{ or } m = -2$$

$$\Downarrow$$

$$\textcircled{B}$$

Q5. $\frac{dv}{dt} = +5 \frac{\text{cm}}{\text{min}}$

$$\frac{dV}{dt} \left[\frac{\text{cm}^3}{\text{min}} \right] = ? , r = 6 \text{ cm}$$

$$V = \frac{4}{3}\pi r^3$$

Sphere

$$\frac{dV}{dt} = \frac{4}{3}\pi 3r^2 \frac{dr}{dt}$$

$$= \frac{4}{3}\pi \times 3 \times 6^2 \times 5 \frac{\text{cm}}{\text{min}} = 720\pi \frac{\text{cm}^3}{\text{min}} \Rightarrow \textcircled{D}$$

Q8. $t = \tan \frac{\theta}{2}$

$4 \sin \theta + 3 \cos \theta + 5 = ?$

$4 \frac{2t}{1+t^2} + 3 \frac{1-t^2}{1+t^2} + 5 =$

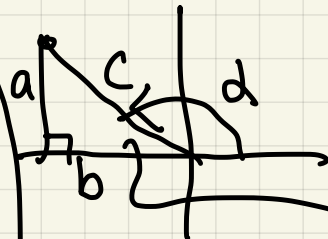
$= \frac{8t + 3 - 3t^2 + 5 + 5t^2}{1+t^2}$

$= \frac{2t^2 + 8t + 8}{1+t^2}$

$= \frac{2(t^2 + 4t + 4)}{1+t^2}$

$= \frac{2(t+2)^2}{1+t^2} \Rightarrow \textcircled{C}$

Q9. $\cos^{-1}(\sin d)$ $\frac{\pi}{2} < d < \pi \Rightarrow$ 2nd quad.



$\sin d > 0$

$\sin d = \sin(180-d)$

$= \frac{a}{c}$

$\therefore \cos^{-1}(\sin d) =$

$= \cos^{-1}\left(\frac{a}{c}\right)$

$= \angle ABC$

Or use test cases:

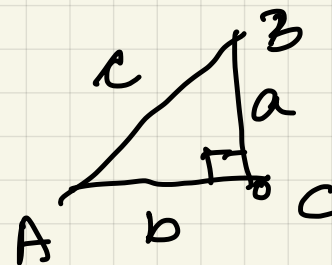
$\cos^{-1}(\sin \frac{5\pi}{6}) = \cos^{-1}(\frac{1}{2})$

$= \frac{\pi}{3}$

if $d = \frac{5\pi}{6}$,

$\pi - d = \frac{\pi}{6}$ not A)

$\frac{\pi}{2} - d = \frac{-2\pi}{6} = \frac{-\pi}{3}$ not B)



$\angle BAC = \pi - d$

$\therefore \angle ABC = \frac{\pi}{2} - (\pi - d)$

$= -\frac{\pi}{2} + d$

$= d + \frac{\pi}{2} \Rightarrow \textcircled{C}$

$\Rightarrow d + \frac{\pi}{2} = \frac{8\pi}{6}$ not D)

$\therefore \textcircled{C}$

$$Q_{10} \quad y = \sin 2x$$

$$I: y = -\sin 2x$$

$$II: y = -\sin 2x - 1$$

$$\begin{aligned} \underline{III}: y &= -\sin(2x \times \frac{1}{2}) - 1 \\ &= -\sin x - 1 \\ &= -(\sin x + 1) \Rightarrow (A) \end{aligned}$$

$$Q_1 \quad C$$

$$Q_2 \quad A$$

$$Q_3 \quad A$$

$$Q_4 \quad C$$

$$Q_5 \quad D$$

$$Q_6 \quad A$$

$$Q_7 \quad B$$

$$Q_8 \quad C$$

$$Q_9 \quad C$$

$$Q_{10} \quad A$$

Q11 a) $\frac{1-2x}{1+x} \geq 1, x \neq -1$

$$\frac{1-2x}{1+x} - 1 \geq 0$$

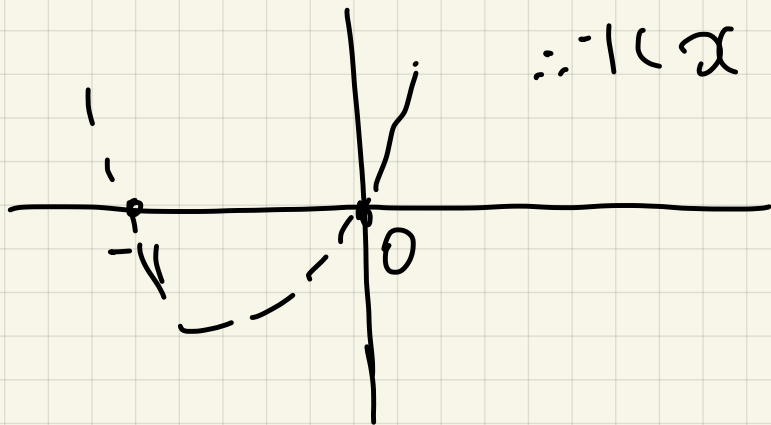
$$\frac{1-2x-1-x}{1+x} \geq 0$$

$$\frac{-3x}{1+x} \geq 0$$

$$\frac{3x}{1+x} \leq 0 \quad | \quad x(1+x)^2$$

$$3x(1+x) \leq 0$$

$\textcircled{1} + \textcircled{1}$
 $\therefore -1 < x \leq 0$



Q11 b)
 i) $\sin 2x + \sqrt{3} \cos 2x$
 $R \sin(2x + \alpha)$

$$R \sin 2x \cos \alpha + R \cos 2x \sin \alpha =$$

$$= \sin 2x + \sqrt{3} \cos 2x$$

$$\therefore R \cos \alpha = 1$$

$$R \sin \alpha = \sqrt{3}$$

$$\left. \begin{array}{l} R \cos \alpha = 1 \\ R \sin \alpha = \sqrt{3} \end{array} \right\} R^2 = 4 \Rightarrow R = 2 \textcircled{1}$$

$$\tan \alpha = \sqrt{3}$$

$$\therefore \alpha = \frac{\pi}{3} \textcircled{1}$$

$$\therefore \sin 2x + \sqrt{3} \cos 2x = 2 \sin(2x + \frac{\pi}{3})$$

ii) greatest value for $2 \sin(2x + \frac{\pi}{3}) = 2$
 $\therefore 2$ for $0 \leq x \leq 2\pi$
 $\textcircled{1}$

$$0 \leq 2\alpha \leq 4\pi$$

$$\frac{\pi}{3} \leq 2\alpha + \frac{\pi}{3} \leq 4\pi + \frac{\pi}{3}$$

$$\frac{\pi}{3} \leq 2\alpha + \frac{\pi}{3} \leq \frac{13\pi}{3}$$

$$\therefore 2\alpha + \frac{\pi}{3} = \frac{\pi}{2}, \frac{5\pi}{2}$$

$$2\alpha = \frac{\pi}{2} - \frac{\pi}{3}, \frac{5\pi}{2} - \frac{\pi}{3}$$

$$2\alpha = \frac{3\pi - 2\pi}{6}, \frac{15\pi - 2\pi}{6}$$

$$\alpha = \frac{\pi}{12}, \frac{13\pi}{12} \quad \textcircled{1}$$

ii) $\sin 2\alpha + \sqrt{3} \cos 2\alpha = 1, 0 \leq \alpha \leq 2\pi$ $\therefore$

$$2 \sin(2\alpha + \frac{\pi}{3}) = 1$$

$$\sin(2\alpha + \frac{\pi}{3}) = \frac{1}{2}$$

$$2\alpha + \frac{\pi}{3} = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$

$$2\pi + \frac{\pi}{6}, 3\pi - \frac{\pi}{6}$$



①

$$2\alpha + \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, 4\pi + \frac{\pi}{6}$$

$$2\alpha = \frac{\pi}{6} - \frac{\pi}{3}, \frac{5\pi}{6} - \frac{\pi}{3}, \frac{13\pi}{6} - \frac{\pi}{3}, \frac{17\pi}{6} - \frac{\pi}{3}$$

$$2\alpha = \frac{-\pi}{6}, \frac{3\pi}{6}, \frac{11\pi}{6}, \frac{15\pi}{6}, \frac{25\pi}{6} - \frac{\pi}{3}$$

$$\alpha = \frac{-\pi}{12}, \frac{3\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{23\pi}{12} \quad \textcircled{1}$$

$$\alpha = \frac{3\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{23\pi}{12} \quad \textcircled{1}$$

Q11 c) $u = \tan^{-1} x$
 $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$

$$du = \frac{dx}{1+x^2} \quad \textcircled{1}$$

$$\int e^u du = e^u + C$$

$$= e^{\tan^{-1} x} + C \quad \textcircled{1}$$

$$\rightarrow P(x) = ax^3 + cx = x(ax^2 + c)$$

$$P(-4) = -4(16a + c) = 0 \Rightarrow c = -16a$$

$$P(3) = 3(9a + c) = 21 \Rightarrow 9a + c = 7$$

$$\therefore 9a - 16a = 7$$

$$-7a = 7$$

$$a = -1 \quad \textcircled{1}$$

$$\therefore c = (-16) \times (-1) = 16 \quad \textcircled{1}$$

$$\therefore P(x) = -1x^3 + 16x$$

$$= -x^3 + 16x$$

$$= 16x - x^3$$

Q11 d) odd $P(-x) = -P(x)$

$$P(x) = ax^3 + bx^2 + cx + d$$

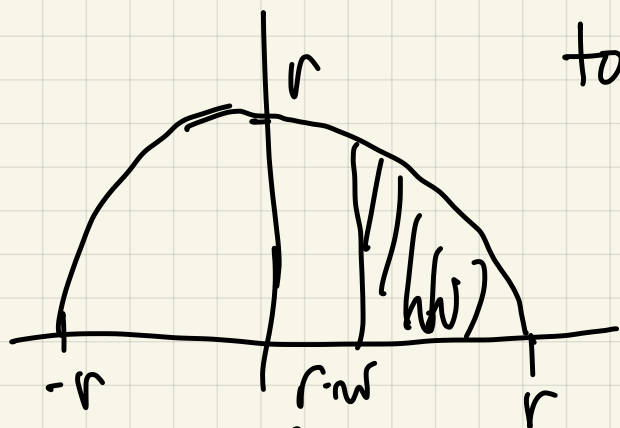
$$P(-x) = a(-x)^3 + b(-x)^2 + c(-x) + d = -ax^3 + bx^2 - cx + d$$

$$b = b \Rightarrow b = 0$$

$$d = -d \Rightarrow d = 0$$

$$\therefore P(x) = ax^3 + cx \quad \textcircled{1}$$

Q2 a) $y = \sqrt{r^2 - x^2}$



to show

$$V = \frac{\pi}{3} (3r-w)w^2$$

$$V = \pi \int_{r-w}^r y^2 dx$$

$$= \pi \int_{r-w}^r (r^2 - x^2) dx \quad \textcircled{1}$$

$$= \pi \left[r^2 x - \frac{x^3}{3} \right]_{r-w}^r =$$

$$= \pi \left[r^2 (r - (r-w)) - \frac{1}{3} (r^3 - (r-w)^3) \right]$$

$$= \pi \left[r^2 w - \frac{1}{3} (r^3 - r^3 + 3r^2 w - 3rw^2 + w^3) \right]$$

$$= \pi \left[rw^2 - \frac{1}{3} w^3 \right]$$

$$= \frac{\pi}{3} w^2 [3r - w]$$

$$\therefore V = \frac{\pi}{3} (3r-w)w^2 \quad \textcircled{1}$$

$$Q12 b) \cos x - 2 \sin x = 1 \quad -\pi \leq x \leq \pi \quad \Rightarrow \quad -\frac{\pi}{2} \leq \frac{x}{2} \leq \frac{\pi}{2}$$

from ref. sheet : $\sin x = \frac{2t}{1+t^2}$

i) $t = \tan \frac{x}{2}$ $\cos x = \frac{1-t^2}{1+t^2}$

ii) $t(t+2) = 0$
 $t = 0$ or $t = -2$

$$\therefore \tan \frac{x}{2} = 0 \text{ or } -2 \quad (1)$$

$$\frac{1-t^2}{1+t^2} - 2 \frac{2t}{1+t^2} = 1$$

$$\frac{1-t^2-4t}{1+t^2} = 1 \quad (1)$$

$$-t^2 - 4t + 1 = 1 + t^2$$

$$-2t^2 - 4t = 0$$

$$2t^2 + 4t = 0$$

$$t^2 + 2t = 0 \quad (1)$$

$$\frac{x}{2} = \arctan 0 \text{ or } \arctan(-2)$$

$$x = 0 \text{ or } 2 \arctan(-2)$$

$$x = 0 \text{ or } -2.2143 \quad (1)$$

Q12c) i) $P(2\sin x + \cos x) + Q(2\cos x - \sin x) = 7\sin x + 11\cos x$

$$2P\sin x + P\cos x + 2Q\cos x - Q\sin x = 7\sin x + 11\cos x$$

$$\therefore \left. \begin{array}{l} 2P - Q = 7 \quad / \times 2 \\ P + 2Q = 11 \end{array} \right\} +$$

$$5P = 25$$

$$P = 5 \Rightarrow Q = 2P - 7 = 10 - 7 = 3$$

$\therefore$ $\textcircled{1}$ $P = 5$, $\textcircled{1}$ $Q = 3$

ii) $\int_0^{\frac{\pi}{2}} \frac{7\sin x + 11\cos x}{2\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{5(2\sin x + \cos x)}{2\sin x + \cos x} dx +$

$$3 \int_0^{\frac{\pi}{2}} \frac{2\cos x - \sin x}{2\sin x + \cos x} dx = 5 \int_0^{\frac{\pi}{2}} dx + 3 \int_0^{\frac{\pi}{2}} \frac{f'(x)}{f(x)} dx =$$

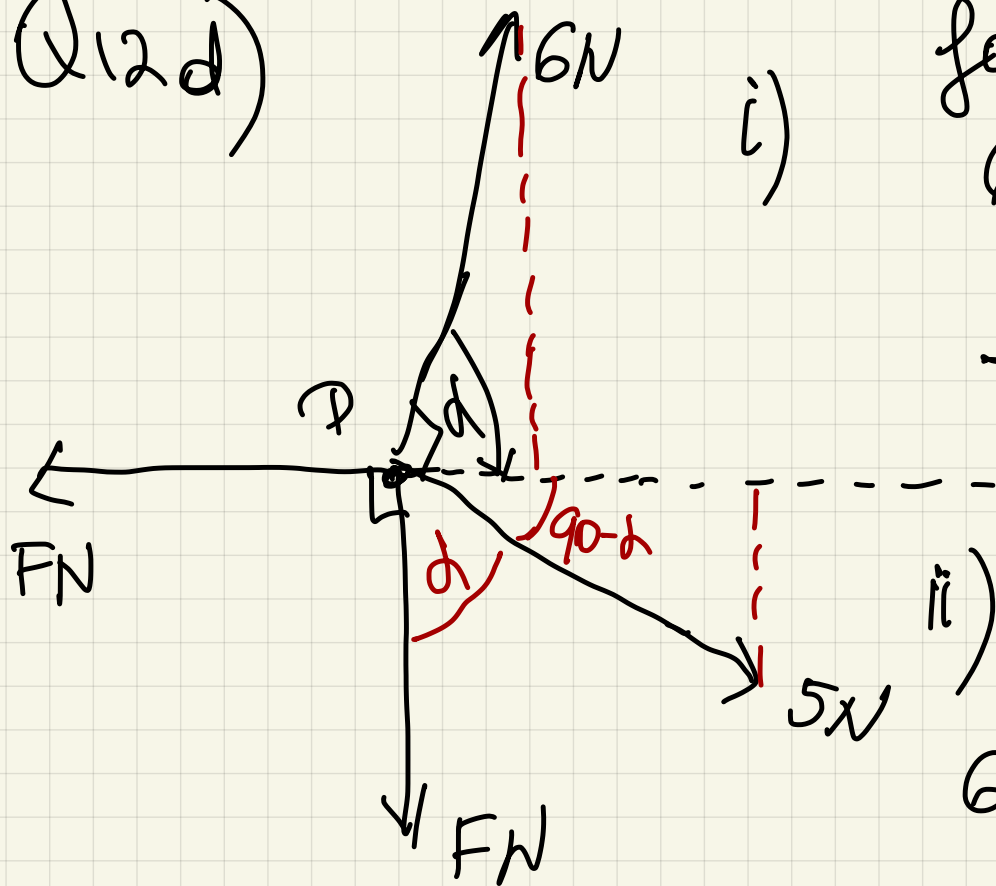
Q2c) ii) continued

$$5 \left(\frac{\pi}{2} - 0 \right) + 3 \ln |2 \sin x + \cos x| \Bigg|_0^{\frac{\pi}{2}}$$

$$= \frac{5\pi}{2} + 3 \left[\ln |2+0| - \ln 1 \right]$$

$$= \frac{5\pi}{2} + 3 \ln 2 \quad \textcircled{1}$$

Q12d)



i)

for vertical components

$$6 \sin \alpha - 5 \sin(90 - \alpha) - F + 0 = 0$$

$$\therefore F = 6 \sin \alpha - 5 \cos \alpha \quad (1)$$

ii) for horizontal components

$$6 \cos \alpha + 5 \cos(90 - \alpha) + 0 - F = 0 \quad (1)$$

$$6 \cos \alpha + 5 \sin \alpha = F \quad \left\{ \begin{array}{l} - \\ \end{array} \right.$$

$$6 \sin \alpha - 5 \cos \alpha = F \quad \left\{ \begin{array}{l} - \\ \end{array} \right.$$

from (i) :

$$11 \cos \alpha - \sin \alpha = 0 \quad (1)$$

$$\tan \alpha = 11$$

$$\alpha = 84.80557 \approx 85^\circ \quad (1)$$

Q13 a)

i) prove $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$

$$\text{let } y = \cos^{-1}(-x)$$

$$\cos y = -x$$

$$\begin{aligned}\cos(\pi - y) &= -\cos y \\ &= -(-x) \\ &= x\end{aligned}$$

①

$$\pi - y = \cos^{-1}(x)$$

$$\therefore \pi - \cos^{-1}(-x) = \cos^{-1}(x)$$

$$\cos^{-1}(-x) = \pi - \cos^{-1}(x)$$

①

$$\begin{aligned}
 \text{ii) } & \cos^{-1}(\sin(-\frac{\pi}{3})) \\
 &= \cos^{-1}(\cos(\frac{\pi}{2} - (-\frac{\pi}{3}))) \text{ using } \cos(\frac{\pi}{2} - x) = \sin x \quad \textcircled{1} \\
 &= \cos^{-1}(\cos \frac{5\pi}{6}) \\
 &= \frac{5\pi}{6} \textcircled{1}, \text{ as } \frac{5\pi}{6} \in [0, \pi]
 \end{aligned}$$

$\leftarrow$ restriction of domain for $\cos^{-1}(\cos(x)) = x, x \in [0, \pi]$

$$\begin{aligned}
 \text{or } & \cos^{-1}(\sin(-\frac{\pi}{3})) \\
 &= \cos^{-1}(-\frac{\sqrt{3}}{2}) \text{ using (i) part } \textcircled{1} \\
 &= \pi - \cos^{-1}(\frac{\sqrt{3}}{2}) \\
 &= \pi - \frac{\pi}{6} \\
 &= \frac{5\pi}{6} \textcircled{1}
 \end{aligned}$$

Q13b) $R = 3\text{m}$, $\frac{dV}{dt} = \frac{\pi}{3} \frac{\text{m}^3}{\text{min}}$

i) to show $V = \frac{\pi}{3} (9h^2 - h^3)$

using Pythagoras

$$3^2 = (3-h)^2 + r^2$$

$$r^2 = 9 - (3-h)^2$$

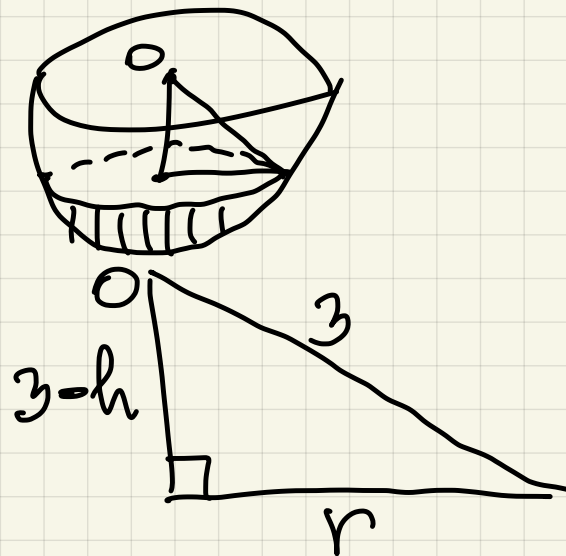
$$= 9 - 9 + 6h - h^2$$

$$= 6h - h^2 \quad \textcircled{1}$$

Volume: $V = \pi \int_0^h (6h - h^2) dh$

$$= \pi \left[6 \frac{h^2}{2} - \frac{h^3}{3} \right]_0^h$$

$$= \pi \left[3h^2 - \frac{h^3}{3} \right] = \frac{\pi}{3} [9h^2 - h^3] \text{ m}^3 \quad \textcircled{1}$$



or

Alternatively:

Rotated $x^2 + y^2 = 9$ around y-axis
between 3 and $3-h$

$$\therefore x^2 = 9 - y^2$$

$$V = \pi \int_{3-h}^3 x^2 dy$$

$$= \pi \int_{3-h}^3 (9 - y^2) dy$$

$$= \pi \left[9y - \frac{y^3}{3} \right]_{3-h}^3$$

$$= \pi \left[27 - \frac{27}{3} - \left((27 - 9h) - \frac{(3-h)^3}{3} \right) \right] =$$

$$\pi \left[\cancel{27} - 9 - \cancel{27} + 9h + \frac{1}{3}(27 - 27h + 9h^2 - h^3) \right]$$

$$= \pi \left[\cancel{-9} + \cancel{9h} + \cancel{9} - \cancel{9h} + 3h^2 - \frac{1}{3}h^3 \right]$$

$$= \pi \left[3h^2 - \frac{1}{3}h^3 \right]$$

$$= \frac{\pi}{3} [9h^2 - h^3]$$

ii) $h = ?$ $t = 8 \text{ min}$

$$\frac{dV}{dt} = \frac{\pi}{3} \frac{m^3}{\text{min}}$$

$$V = \frac{\pi}{3} t \frac{m^3}{\text{min}}, \text{ at } t = 8 \text{ min}$$

$$\Rightarrow V = \frac{8\pi}{3} \frac{m^3}{\text{min}} \quad (1)$$

$$\frac{8\pi}{3} = \frac{\pi}{3} (9h^2 - h^3)$$

$$8 = 9h^2 - h^3$$

$$h^3 - 9h^2 + 8 = 0$$

$$(h-1)(h^2 - 8h - 8) = 0 \quad (1)$$

$$h-1=0 \text{ or } h^2 - 8h - 8 = 0$$

$$\therefore h=1 \text{ or } h = \frac{8 \pm \sqrt{64 - 4 \times (-8) \times 1}}{2} = \frac{8 \pm \sqrt{64 + 32}}{2} = \frac{8 \pm \sqrt{96}}{2}$$

$$h^3 - 9h^2 + 8 = (h-1)(h^2 - 8h - 8)$$

1	1	-9	0	8
		1	-8	-8
	1	-8	-8	0

$$= \frac{8 \pm \sqrt{8 \times 12}}{2}$$

$$= \frac{8 \pm \sqrt{8 \times 2 \times 6}}{2}$$

$$= \frac{8 \pm 4\sqrt{6}}{2}$$

$$= 4 \pm 2\sqrt{6} \quad \text{as}$$

$$0 \leq h \leq 3$$

$$\therefore h=1 \text{ only} \quad (1)$$

$$\text{iii) } \frac{dh}{dt} = ?$$

$$\frac{dV}{dt} = \frac{\pi}{3} \text{ (given)}, V = \frac{\pi}{3} [9h^2 - h^3] \text{ (proved)}$$

$$\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt}$$

$$\frac{\pi}{3} = \frac{dV}{dh} \times \frac{dh}{dt} \quad (1)$$

$$\frac{\pi}{3} = \pi [6h - h^2] \times \frac{dh}{dt}$$

$$\frac{dV}{dh} = \frac{\pi}{3} [18h - 3h^2] \\ = \pi [6h - h^2]$$

$$\Rightarrow \frac{dh}{dt} = \frac{\pi}{3} \times \frac{1}{\pi [6h - h^2]}$$

$$\frac{dh}{dt} = \frac{1}{3 [6h - h^2]}$$

$$\text{when } h=1 \text{ m} \Rightarrow \frac{dh}{dt} =$$

$$\therefore \frac{dh}{dt} = \frac{1}{3[6h-h^2]}$$

for $h=1$ m

$$\frac{dh}{dt} = \frac{1}{3[6 \times 1 - 1]} = \frac{1}{3 \times 5} = \frac{1}{15} \frac{\text{m}}{\text{min}}$$

①

$\therefore$ when oil is 1 m deep, h is increasing at a rate of $\frac{1}{15}$ meters per minute.

$$Q13 C) \sin q + \sin 3q + \sin 5q + \dots + \sin (2n-1)q = \frac{1 - \cos 2nq}{2 \sin q}$$

Step 1: $n=1$

$$LHS = \sin q$$

$$RHS = \frac{1 - \cos 2q}{2 \sin q}$$

$$= \frac{1 - (1 - 2 \sin^2 q)}{2 \sin q}$$

$$= \frac{2 \sin^2 q}{2 \sin q}$$

$$= \sin q = LHS$$

$\therefore$ true for $n=1$

①

Step 2:

$$\text{Assume true } n=k : \sin q + \sin 3q + \sin 5q + \dots + \sin (2k-1)q = \frac{1 - \cos 2kq}{2 \sin q}$$

To prove true for $n=k+1$:

$$\sin q + \sin 3q + \sin 5q + \dots + \sin (2k+1)q = \frac{1 - \cos 2(k+1)q}{2 \sin q}$$

$$\begin{aligned}
\text{LHS} &= \sin q + \sin 3q + \sin 5q + \dots + \sin(2k-1)q + \sin(2k+1)q \\
&= \frac{1 - \cos 2kq}{2 \sin q} + \sin(2k+1)q \quad \text{(by assumption)} \\
&= \frac{1 - \cos 2kq + 2 \sin q \sin(2k+1)q}{2 \sin q} \\
&= \frac{1 - \cos [(2k+1)q - q] + 2 \sin q \sin(2k+1)q}{2 \sin q} \quad \textcircled{1} \\
&= \frac{1 - [\cos(2k+1)q \cos q + \sin(2k+1)q \sin q] + 2 \sin q \sin(2k+1)q}{2 \sin q} \\
&= \frac{1 - [\cos(2k+1)q \cos q - \sin(2k+1)q \sin q]}{2 \sin q} \\
&= \frac{1 - \cos[(2k+1)q + q]}{2 \sin q} = \frac{1 - \cos(2k+2)q}{2 \sin q} \quad \textcircled{1}
\end{aligned}$$

$$= \frac{1 - \cos 2(k+1)q}{2 \sin q}$$

Step 3: $\hookrightarrow$ If statement true for $n=k$ it is true for $n=k+1$.
Since true for $n=1, \dots, n=k$, and $n=k+1$, it is
true for any n . ①

Q14a)

$$f(x) = 2 - \sqrt{x}, x \geq 0$$

$$g(x) = (x-2)^2, \text{ for all } x$$

i) $f(g(x)) = g(f(x))$

Domain: $[0, +\infty)$

Range: $(-\infty, 2]$

$$f(g(x)) = f((x-2)^2) = 2 - \sqrt{(x-2)^2}$$

$$\sqrt{(x-2)^2} = \begin{cases} x-2, & x \geq 2 \\ -x+2, & x < 2 \end{cases}$$

①

$$\therefore f(g(x)) = \begin{cases} 2 - (x-2) = 4-x, & x \geq 2 \\ 2 - (-x+2) = x, & x < 2 \end{cases} = \begin{cases} 4-x, & x > 2 \\ 4-2=2=x, & x=2 \\ x, & x < 2 \end{cases}$$

$$f(f(x)) = f(2 - \sqrt{x}) = ((2 - \sqrt{x}) - 2)^2 = (-\sqrt{x})^2 = x$$

①

$$\therefore [0, +\infty) \cap (-\infty, 2] = [0, 2]$$

for all $x \geq 0$

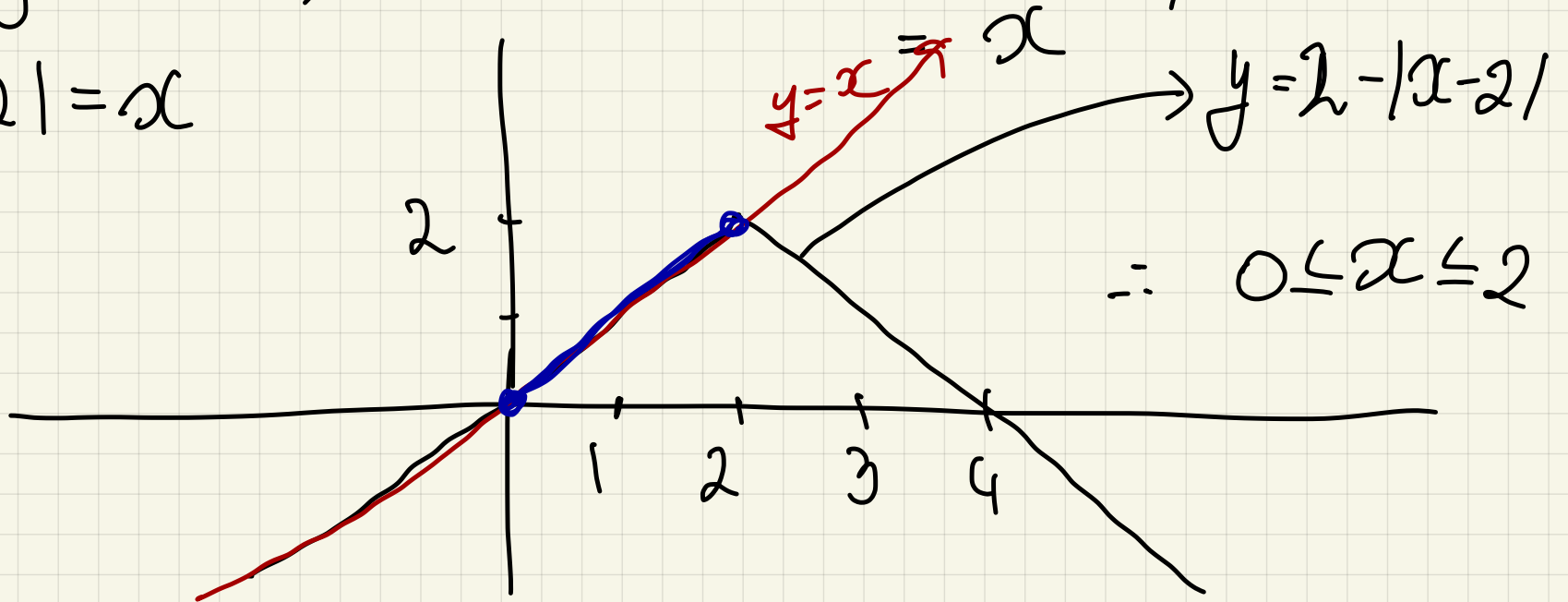
$$\therefore f(g(x)) = x = g(f(x)) \text{ for } 0 \leq x \leq 2 \quad \textcircled{1}$$

Alt:

$$f(g(x)) = f((x-2)^2) = 2 - \sqrt{(x-2)^2} = 2 - |x-2|$$

$$g(f(x)) = g(2 - \sqrt{x}) = (2 - \sqrt{x} - 2)^2 = |x|, \text{ but } x \geq 0$$

$$2 - |x-2| = x$$



$$Q14 b) x = vt \cos \alpha \quad (1)$$

$$y = -\frac{1}{2}gt^2 + vt \sin \alpha \quad (2)$$

$$i) v \sin \alpha = \sqrt{2gh} \quad \text{to show}$$

$$\dot{y} = -\frac{1}{2} \times 2gt + v \sin \alpha = -gt + v \sin \alpha$$

$$\dot{y} = 0 \quad \text{for maximum height}$$

$$0 = -gt + v \sin \alpha$$

$$gt = v \sin \alpha$$

$$t = \frac{v \sin \alpha}{g} \quad \text{into (2) for } y_{\max} = 3h \quad \longrightarrow \textcircled{1}$$

$$y_{\max} = -\frac{1}{2}g\left(\frac{v \sin \alpha}{g}\right)^2 + v \frac{v \sin \alpha}{g} \sin \alpha = -\frac{1}{2} \frac{v^2 \sin^2 \alpha}{g} + \frac{v^2 \sin^2 \alpha}{g} =$$

$$y_{\max} = \frac{1}{2} \frac{v^2 \sin^2 \theta}{g}$$

$$y_{\max} = 3h \text{ (given)}$$

$$3h = \frac{1}{2} \frac{v^2 \sin^2 \theta}{g}$$

$$6gh = v^2 \sin^2 \theta \quad (1)$$

$$\therefore v \sin \theta = \sqrt{6gh}$$

$$\text{ii) } v \cos \theta = \frac{gd}{2\sqrt{6gh}} \quad \text{to show}$$

$$\text{Range} = d, \text{ where } y = 0$$

$$0 = t(v \sin \theta - \frac{1}{2}gt)$$

$$t = 0 \text{ or } v \sin \theta - \frac{1}{2}gt = 0$$

at the start at the end of flight

$$\therefore \frac{1}{2}gt = v \sin \theta$$

$$t = \frac{2v \sin \theta}{g} \text{ time of flight}$$

so

$$\text{Range} = vt \cos \theta$$

$$= v \times \frac{2v \sin \theta}{g} \times \cos \theta$$

$$\text{Range} = \frac{v^2}{g} 2 \sin \theta \cos \theta$$

$$\text{Range} = d$$

$$\frac{v^2}{g} 2 \sin \alpha \cos \alpha = d$$

$$\underline{v \cos \alpha} \times 2 \frac{v}{g} \sin \alpha = d$$

$$v \cos \alpha = d \times \frac{1}{2} \frac{g}{v \sin \alpha}$$

$$\text{now } v \sin \alpha = \sqrt{6gh}$$

$$\therefore v \cos \alpha = \frac{gd}{2} \times \frac{1}{\sqrt{6gh}} \quad (1)$$

$$= \frac{gd}{2\sqrt{6gh}}$$

iii) to show $y = \frac{12hx}{d} \left(1 - \frac{x}{d}\right)$

$$t = \frac{x}{v \cos \alpha} \quad \text{Sub}$$

$$y = -\frac{1}{2}g \left(\frac{x^2}{v^2 \cos^2 \alpha} \right) + v \frac{x}{v \cos \alpha} \sin \alpha \quad (1)$$

$$= -\frac{1}{2}g \frac{x^2}{\frac{24gh}{2}} + 12 \frac{h}{d} x$$

$$\frac{v \sin \alpha}{v \cos \alpha} = \frac{\sqrt{6gh}}{\frac{gd}{2\sqrt{6gh}}}$$

$$= \frac{12gh}{gd}$$

$$= 12 \frac{h}{d} \quad (*)$$

$$\therefore y = -12 \frac{hx^2}{d^2} + 12 \frac{h}{d} x$$

$$= \frac{12hx}{d} \left(1 - \frac{x}{d}\right) \quad \textcircled{1}$$

$$iv) \sqrt{x} = 100(3 + \sqrt{6}) \frac{u}{s} \Rightarrow x = v_a t$$

$$t = f(d) = ?$$

at Q $y = h$

from (iii) $y = \frac{12hx}{d} \left(1 - \frac{x}{d}\right)$

$$h = \frac{12hx}{d} \left(1 - \frac{x}{d}\right)$$

$$1 = 12 \frac{x}{d} \left(1 - \frac{x}{d} \right)$$

$$d^2 = 12xd \left(1 - \frac{x}{d} \right)$$

$$d^2 = 12xd - 12x^2$$

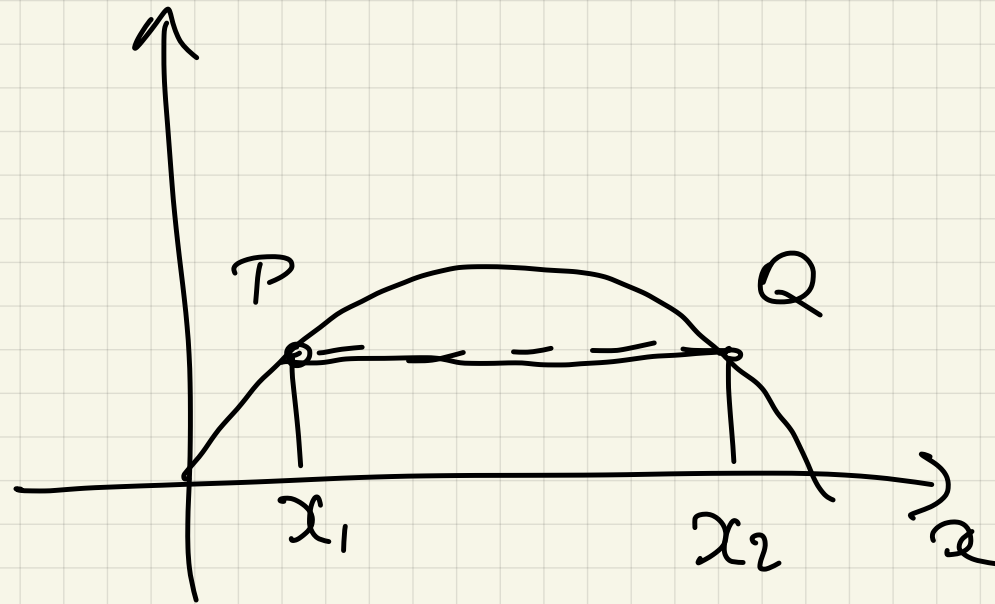
$$12x^2 - 12xd + d^2 = 0 \quad (1)$$

$$x_{1/2} = \frac{12d \pm \sqrt{144d^2 - 4 \times 12 \times d^2}}{24}$$

$$= \frac{12d \pm \sqrt{96d^2}}{24}$$

$$= \frac{d}{2} \pm \frac{4d\sqrt{6}}{24}$$

$$= \frac{d}{2} \pm \frac{d\sqrt{6}}{6} = \frac{d(3 \pm \sqrt{6})}{6}$$



$$\rightarrow \text{at } Q: x = \frac{3+\sqrt{6}}{6}d$$

$$x = \sqrt{x}t$$

$$\frac{3+\sqrt{6}}{6}d = 100(3+\sqrt{6})t$$

$$\therefore t = \frac{d}{600} \quad (1)$$

v) $u = ?$ just horizontal component $\Rightarrow x = ut$

$$PQ = \text{distance} = x_2 - x_1 = \frac{3+\sqrt{6}}{6}d - \frac{3-\sqrt{6}}{6}d$$

$$\therefore PQ = \frac{2\sqrt{6}}{6}d = \frac{\sqrt{6}}{3}d$$

$$\therefore \frac{\sqrt{6}}{3}d = u \times \frac{d}{600}$$

$$\therefore u = \frac{\frac{\sqrt{6}}{3}d}{\frac{d}{600}} = \frac{600\sqrt{6}}{3} = 200\sqrt{6}$$

$$u \approx 490 \frac{\text{m}}{\text{s}} \quad \textcircled{1}$$

Q 14 c)

$b_k = a_{k-1} + a_k$ from Pascal Triangle

$$\frac{1}{a_0 a_1 a_2 \dots a_n} \times (a_0 + a_1)(a_1 + a_2)(a_2 + a_3) \dots (a_{n-1} + a_n)$$

$n = 1, 2, 3, \dots$

$$= \frac{1}{a_0} \times \frac{1}{a_1 a_2 \dots a_n} \times b_1 b_2 b_3 \dots b_n, \quad a_0 = 1$$

$$= 1 \times \frac{b_1 b_2 \dots b_n}{a_1 a_2 \dots a_n}$$

now

$$\frac{b_k}{a_k} = \frac{{}^{n+1}C_k}{{}^n C_k} = \frac{(n+1)!}{(n+1-k)! k!} \times \frac{(n-k)! k!}{n!}$$

$$\frac{b_k}{a_k} = \frac{n+1}{n-k+1}$$

$$\frac{b_1 b_2 \dots b_n}{a_1 a_2 \dots a_n} = \frac{(n+1)(n+1) \dots (n+1)}{n(n-1)(n-2) \dots 1}$$

$$= \frac{(n+1)^n}{n!}$$

①

$$\therefore \frac{1}{a_0 a_1 a_2 \dots a_n} \times (a_0 + a_1)(a_1 + a_2) \dots (a_{n-1} + a_n)$$

$$= \frac{(n+1)^n}{n!}$$